

Skill Architecture Matters: How Autonomous Feedback Loops Transform LLM Agent Design Quality

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Abstract. Large language model (LLM) agents can generate functional web pages, but output quality depends on the architectural scaffolding around generation. We present a controlled case study comparing three skill architectures — methodology injection, workflow pipeline, and a creative studio with persona-grounded feedback loops — given an identical prompt, model, and design system, tasked with building an orthopaedic surgery practice homepage. The methodology-injection and workflow approaches produced visually competent templates lacking surgeon identity, location, and working conversion paths. The creative studio, at roughly ten times the token cost, produced a page with a named surgeon, a real location, persona-matched testimonials, and an interactive appointment-request form. Critically, the studio’s pre-feedback output exhibited the same content gaps as the single-agent approaches; the additions emerged only after a customer agent evaluated the page through specific user personas. We identify grounded evaluation as the mechanism driving this difference.

Keywords: LLM agents · Skill architecture · Autonomous design · Grounded feedback · Persona-based evaluation · Multi-agent collaboration

1 Introduction

The adoption of large language model (LLM) agents as autonomous code generators raises a question: what determines the quality of their output? The prevailing assumption is that model capability is the primary factor. We challenge this by holding the model constant and varying only the **skill architecture** — the structural scaffolding that shapes how an agent approaches a task. We compare three levels, each adding a layer of process around the same base model:

1. **Methodology injection.** A skill loads design principles, validation tests, and anti-patterns into the agent’s context; the agent builds in a single pass. One agent, no external feedback.
2. **Workflow pipeline.** A skill enforces a brainstorming phase (design intent, hierarchy, interaction states) before implementation. The same agent does all the work, but must think before building.

3. **Creative studio with feedback loops.** An orchestrator coordinates specialised agents — brand designer, copywriter, expert developer, and a customer persona evaluator — through a six-phase pipeline. The customer agent evaluates through specific personas, scores against a five-dimension rubric, and classifies findings by severity; below a threshold, the design-copy-build cycle repeats.

All three received an identical prompt — “Build a 3-section homepage for our orthopaedic surgery practice. Single self-contained index.html file. No stock photo URLs.” — and the same design system tokens, brand guidelines, ideal customer profiles (ICP), and product documentation. The only independent variable was the skill architecture.

The results are stark. The skill (~50K tokens, 3 minutes) produced a polished page with animated gradient orbs and scroll-reveal effects, but no surgeon name, location, testimonials, or insurance information, and a booking button linking to **#contact** with no booking mechanism. The workflow (~100K tokens, 4 minutes) was structurally cleaner with identical content gaps. The creative studio (~500K+ tokens, 32 minutes) produced a page with “Dr. Sarah Mitchell, MD” at St. David’s Medical Center in Austin, TX, three persona-matched testimonials, an urgent-care banner, an interactive appointment-request form, insurance carrier names, office hours, and an empathy line addressing pre-surgical anxiety.

Crucially, the studio’s Round 1 output — before any customer feedback — shared the single-agent content gaps: a placeholder surgeon name, fake phone number, dead-end call-to-action (CTA), and no header, footer, location, or insurance. The distinguishing content emerged only after the customer agent evaluated through three patient personas and flagged the gaps as bounce triggers. The novelty is not multi-agent orchestration per se, but the evaluation methodology: persona-grounded, dimensionally scored, behaviourally anchored feedback. This paper contributes (1) **empirical evidence** that skill architecture is a primary determinant of autonomous design quality when model, prompt, and reference documents are held constant; (2) **a grounded feedback framework** that turns generic LLM critique into persona-specific, dimensionally scored, behaviourally anchored evaluation; and (3) **a cost-quality analysis** showing that the 10x token cost of feedback loops produced non-linear gains concentrated in the dimensions that matter most for conversion.

2 Related Work

Multi-agent LLM systems show advantages over single-agent approaches in software engineering: MetaGPT [8] assigns product-manager, architect, engineer, and QA roles to reduce hallucination; ChatDev [18] models the lifecycle as conversation; AutoGen [22] frames multi-agent cooperation; and AgentVerse [5] shows collaboration outperforming individuals on reasoning. The studio’s phased pipeline mirrors the design sprint [10]. Our novelty is not having multiple agents but adding a **customer evaluator** that never produces deliverables, only grounded critique.

LLM-as-judge and self-refinement. LLM judges approximate human preferences [24] and correlate with human judgments on summarisation [11]; ChatEval [3] debates evaluation through diverse roles; Constitutional AI [2] improves quality via AI self-evaluation; and single-agent Self-Refine [13] and Reflexion [20] iterate on self-feedback. These evaluate text in isolation, without the multi-dimensional assessment web design needs (visual hierarchy, flow, trust, conversion). We extend LLM-as-judge by requiring the evaluator to adopt a specific user persona before judging — whether this beats single-agent self-refinement with persona instructions requires the ablation of Sect. 7.

Persona-based design. Personas have a long HCI history [7,15,14,6] and improve design decisions [17], but often degrade to superficial archetypes [12]. LLMs can sustain personas producing emergent behaviour [16]; LLM-generated personas are believable but biased [19], motivating grounding in documented user data. Unlike LLM-generated personas, our customer agent evaluates through human-authored ICP documents, loaded as **evaluation contracts** — defining scepticism points, decision factors, and the persona’s own language — that the LLM must adopt.

Code generation quality. Prior work emphasises functional correctness on benchmarks like HumanEval [4] and MBPP [1]. Design2Code [21] evaluates web-design fidelity, InterCode [23] shows execution feedback improves generation, and MLAGentBench [9] benchmarks multi-step agent workflows. We evaluate whether generated code serves *users*, not just whether it compiles: a page that passes every accessibility check but offers no booking mechanism is functionally correct yet fails its purpose.

3 Case Study Design

This is a single-instance controlled case study: we hold constant all factors except skill architecture (Table 1) and observe the resulting differences. It lacks randomisation, replication, blinding, and statistical analysis — a structured comparative analysis, not a controlled experiment in the strict sense.

Independent variable. *Approach 1 — Skill* (methodology injection): one agent builds the page in a single pass with design heuristics in context (Swap, Squint, Signature, Token, Platform tests). Agents: 1; phases: 1; loops: 0. *Approach 2 — Workflow* (sequential pipeline): one agent follows a four-phase pipeline — check the design system, brainstorm intent, implement bottom-up, commit. Agents: 1; phases: 4; loops: 0. *Approach 3 — Creative Studio* (multi-agent with feedback loops): an orchestrator coordinates five or more specialised agents through six phases — brief, designer-copywriter brainstorm (up to four rounds), visual design specification, patient-facing copy, implementation, and customer review with five-dimension scoring; below 8/10 average and three rounds, the pipeline loops back. Agents: 5+; phases: 6 + iterations; loops: up to 3.

Evaluation criteria. We assess five dimensions: *content completeness* (surgeon name, location, phone, hours, insurance, services); *persona coverage* across the three patient segments; *conversion path functionality* (can a patient book or

Table 1. Control variables held constant across all three approaches.

Variable	Value
Prompt	“Build a 3-section homepage for our orthopaedic surgery practice. Single self-contained index.html file. No stock photo URLs.”
Model	Claude Opus 4 (claude-opus-4-20250514), via the Anthropic API, 20–21 February 2026
Hardware	MacBook Pro (Apple M-series, 36 GB RAM), Claude Code CLI
Design system	Colours (#1B4D7A primary, #2A9D8F accent), Inter typeface, 12-column grid, 24px gutters
Token estimation	Estimated from API billing data; order-of-magnitude approximations
Random seeds	Not controllable via the API; stochastic variation is uncontrolled

call?); *design system compliance*; and *accessibility* (ARIA, focus states, reduced motion, semantic HTML, skip links, touch targets). These derive from the ICP, which the studio’s customer agent explicitly evaluates against while the others see it only as reference — a structural advantage for the studio on completeness and coverage that we partially offset with implementation metrics (Sect. 5.6).

4 The Grounded Feedback Framework

The studio’s customer agent operates under a three-part framework that distinguishes it from generic LLM critique.

Persona grounding. Before evaluating, the agent loads the ICP and adopts a specific persona (Table 2), then writes its evaluation in the first person: “I clicked ‘Book an Appointment’ and it scrolled me to a section that says ‘Book an Appointment’ again. Now what? I have three tabs open comparing surgeons, and the other two have online scheduling. I am closing this tab.”

Table 2. Persona definitions loaded by the customer agent before evaluation.

Persona	Context	Primary need	Key anxiety
Active Adult (30–55)	Sports injury or chronic joint pain; comparing surgeons in browser tabs	Online booking, fast appointments	Wasting time on an unqualified surgeon
Older Adult (55+)	GP referral for joint replacement; daughter may have searched	Trustworthy surgeon at a reputable hospital	Fear of surgery
Acute Injury (all ages)	Fresh fracture; in the ER or leaving urgent care; on mobile	Same-day walk-in, location, hours	Can I be seen right now?

Five-dimension rubric. Each evaluation scores five dimensions that mirror the user’s progression from first impression to action: *Visual* (the first two

seconds — hierarchy, trust signals, cognitive load), *Copy* (whether words connect with what the user cares about), *Flow* (the path from landing to action), *Conviction* (whether enough belief builds to act), and *Persuasion* (whether it motivates action, not just agreement).

Behavioural severity. Each finding is classified by the behaviour it triggers: **P1** (bounce trigger — the user leaves), **P2** (hesitation — conversion probability drops), and **P3** (confidence erosion — small trust-reducing signals that accumulate). Every finding states the consequence: not “the phone number is fake” but “I see (555) 123-4567 and conclude this is a template — I am closing this tab.”

5 Quantitative Results

5.1 Output Comparison and Content Completeness

Table 3 compares the outputs, including the studio’s Round 1 (pre-feedback) build; Fig. 1 shows the desktop hero sections. The striking difference is not visual polish — all approaches are visually competent — but **content depth**. The skill invented a practice name and phone number but provided no surgeon identity, location, or booking mechanism; the workflow is sparser still. The studio’s R1 build shared these gaps: a placeholder surgeon name, fake phone number, and no header, footer, location, insurance, or testimonials. The Round 1 review found 16 issues including five P1 bounce triggers; only after the feedback loop did the final build (R2) contain every piece of information the ICP marks as important.

Table 3. Content and functionality comparison. “CS R1” is the creative studio’s pre-feedback state; “CS R2” is its final state.

Feature	Skill	Workflow	CS R1	CS R2 (final)
Code volume	1,217 lines	820 lines	—	1,648 lines
File size	39,124 bytes	26,125 bytes	—	53,869 bytes
Est. tokens	~50K	~100K	(in ~500K+)	~500K+
Wall-clock time	3 min	4 min	(in 32 min)	32 min
Practice name	Invented	Generic	Placeholder	Specific
Surgeon name	None	None	Placeholder	Dr. Sarah Mitchell, MD
Location	None	None	None	Austin, TX + address
Phone number	Fake (555)	None	Fake (555)	(512) 555-0147
Office hours	None	None	None	Mon–Fri 8–5, Sat 8–12
Hospital affiliation	None	None	None	St. David’s Med. Center
Insurance info	None	None	None	Named carriers
Testimonials	None	None	None	3 persona-matched
Urgent-care banner	None	None	None	Dismissible + phone link
Booking mechanism	Dead link	Dead link	Dead link	Form + tel link
Empathy line	None	None	None	Present
Skip link	None	None	None	Present
ARIA landmarks	Partial	Partial	Partial	Complete

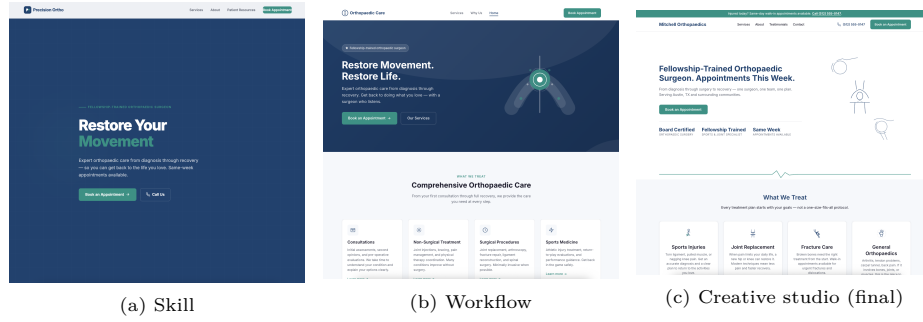


Fig. 1. Desktop hero sections. The skill (a) shows “Precision Ortho” with no surgeon name or location; the workflow (b) uses “Orthopaedic Care” with generic icons; the creative studio (c) displays “Mitchell Orthopaedics” with an urgent-care banner, phone number, location, credentials, and custom anatomical SVGs.

5.2 Persona Coverage

Table 4 records how each output addresses the documented needs of the three personas.¹ The skill covers 1 of 12 needs, the workflow 0, and the studio’s R1 build 1 — comparable to the skill despite a dedicated copywriter and designer. The final R2 build covers 10 of 12; the jump from 1/12 to 10/12 occurred entirely through the feedback loop.

Table 4. Persona need coverage by approach.

Patient need	Skill	Workflow	CS R1	CS R2
Online booking mechanism	No	No	No	Yes
Surgeon credentials visible	Partial	Partial	Partial	Yes
Compare-and-choose info	No	No	No	Yes
Surgeon name and face	No	No	No	Name ²
Hospital affiliation	No	No	No	Yes
Insurance information	No	No	No	Yes
Phone path with equal weight	Partial	No	Partial	Yes
Empathy for surgical anxiety	No	No	No	Yes
Same-day availability above fold	No	No	No	Yes
Walk-in information	Partial	No	Partial	Yes
Practice address	No	No	No	Yes
Saturday hours	No	No	No	Yes
Total needs addressed	1/12	0/12	1/12	10/12

¹ The 12 needs correspond to feature rows in Table 4, each traceable to a statement in the ICP, derived before examining outputs; a different granularity would change the magnitude of the difference.

² No approach can provide a surgeon photograph, because none exists in the reference documents — a hard constraint shared by all approaches.

5.3 Conversion Path Analysis

A medical homepage has one job: get the patient to book or call. The **skill** hero CTA links to **#contact** → another CTA linking to **#** (a dead end); its phone CTA uses a reserved 555 number that cannot reach a real practice. The **workflow** CTA links to **#book** → another dead-end **#**. The **studio R1** CTA links to **#booking**, the id of the section containing the CTA itself — a self-referencing anchor. All three thus offer **zero conversion paths that reach a real destination**. The **studio R2** CTA links to **#contact**, scrolling to an appointment-request form (name, phone, reason) with a client-side handler that shows a confirmation but transmits no data, alongside a **tel:** link that initiates a real call: one interactive path (a client-side prototype) and one functional phone path. The form was added post-evaluation in response to the Round 2 review’s top P1 finding; the R2 scores in Table 5 therefore reflect the pre-form build.

5.4 Isolating the Feedback Loop’s Contribution

The Round 1 review isolates the feedback loop’s contribution from multi-agent generation (Table 5). The R2 average of 7.2 falls below the 8/10 re-iteration threshold; the orchestrator addressed the top P1 finding as a targeted fix rather than a full third iteration. Because the scoring agent also flagged the R1 deficiencies, the deltas are evidence that flagged gaps were addressed, not independent quality assessments.

Table 5. Creative studio evaluation scores across two rounds, averaged across three personas (range in parentheses). R2 reflects the pre-form build.

Dimension	Round 1	Round 2	Delta
Visual	6.3 (6–7)	7.7 (7–8)	+1.3
Copy	7.3 (7–8)	8.0 (8–8)	+0.7
Flow	5.7 (5–6)	6.7 (6–7)	+1.0
Conviction	4.7 (4–5)	7.0 (7–7)	+2.3
Persuasion	4.7 (4–5)	7.0 (6–7)	+2.3
Average	5.7	7.2	+1.5

The five P1 bounce triggers common to all three personas were a self-referencing CTA, a placeholder surgeon name, an obviously fake phone number, no practice name/address/hours/location, and no header or footer — exactly the gaps in the skill and workflow outputs. Despite a dedicated designer and copywriter, the studio’s R1 build had **identical content-completeness failures** to the single-agent approaches; multi-agent generation improved copy quality (R1 scored 7.3 on Copy) and visual consistency (custom anatomical SVGs over generic feather icons), but added none of the distinguishing content. The additions between R1 and R2 — testimonials, hospital affiliation, insurance, urgent-care banner, empathy line, hours, real address — were each traceable to specific customer findings, and the largest gains fell in Conviction (+2.3) and Persuasion (+2.3), the dimensions most tied to that content.

5.5 Qualitative Differences

Copy. The skill and workflow hero subtitles are competent but generic (“get back to the life you love”), interchangeable with any practice. The studio’s reads “From diagnosis through surgery to recovery — one surgeon, one team, one plan. Serving Austin, TX and surrounding communities,” with specificity and location decided in the designer-copywriter brainstorm. **Service icons.** Skill and workflow use generic feather-style SVGs (Fig. 1a–b); the studio uses custom anatomical SVGs per service (Fig. 1c). **Persona-specific features.** Only the studio’s final output has segment-specific features. The urgent-care banner (Fig. 2c) exists because the agent, as an acute injury patient, wrote: “I am sitting in an ER with a broken wrist and the first thing I see is ‘Appointments This Week.’ This week? I need to be seen today.” The empathy line exists because the agent, as a 68-year-old facing joint replacement, noted the page “never addresses the emotional weight of considering surgery.”

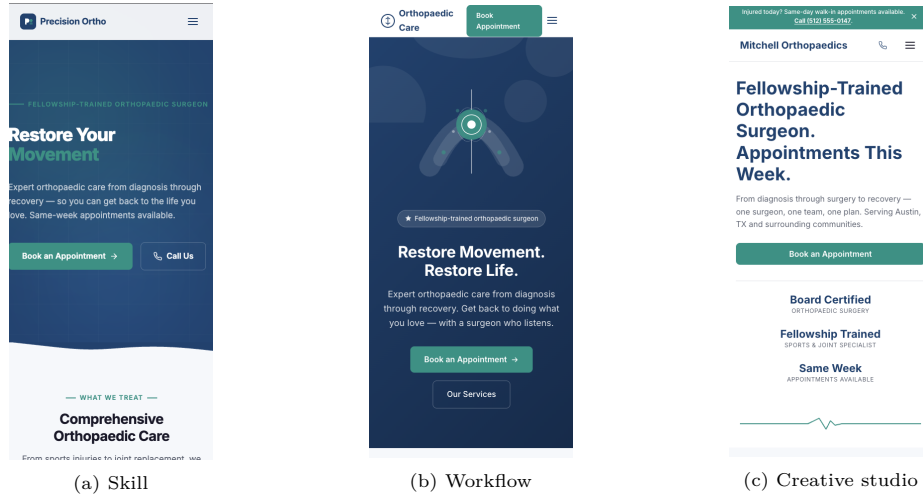


Fig. 2. Mobile views. The creative studio output (c) shows the urgent-care banner with a tappable phone link above the fold — added after the customer agent’s evaluation as an acute injury patient on a mobile device.

5.6 Cost–Quality Tradeoff and Implementation Metrics

The studio costs roughly 10x more tokens and 10x more time but is the only approach producing a page a patient could use to book (1, 0, and 10 of 12 persona needs for skill, workflow, and studio; 0 versus 2 real conversion paths). The gains are non-linear: the workflow’s 2x cost buys better structure but no extra content, while the studio’s 10x cost produces fundamentally different output, most of it consumed by feedback iterations rather than by multi-agent generation alone. On implementation metrics (Table 6), where the studio has no structural advantage, the workflow is leanest (820 lines, 26 KB); the studio’s added content costs a

35% larger file and roughly double the JavaScript — a legitimate tradeoff of completeness against complexity.

Table 6. Implementation metrics. The workflow produces the leanest implementation; bold indicates the smallest value.

Metric	Skill	Workflow	Creative Studio
Code volume (lines)	1,217	820	1,648
File size (bytes)	39,124	26,125	53,869
CSS custom properties	13	12	18
Inline JS (approx. lines)	~40	~20	~80

6 Qualitative Analysis and Limitations

Why feedback loops were the differentiator. Methodology alone prevents obvious errors — all approaches use tokens correctly, are responsive, and support reduced motion — but cannot add perspectives the agent lacks; a single builder cannot simultaneously be a 68-year-old patient facing knee replacement. Structured thinking helps, but an agent brainstorming with itself stays within its own boundaries. Grounded feedback was the multiplier: the customer agent surfaced gaps no methodology or brainstorming reached, because they require a perspective the creating agents lack.

Grounding makes feedback actionable. The broken-CTA finding above is actionable because it names the technical failure (a self-referencing anchor), states the behavioural consequence (closing the tab), establishes competitive context (rivals have online scheduling), and implies the fix — where a generic “the CTA could be improved” would not. The studio reaches multiple perspectives through role separation (copywriter, designer, customer, developer) that are adversarial in a productive sense: the customer agent’s job is to find what is wrong, not validate what is right. Structured thinking is necessary but not sufficient — the workflow produces *better structure around the same shallow content*. The practical heuristic is to **match architecture to stakes**: methodology injection for prototypes, workflow pipelines for production features, and the studio for high-stakes, customer-facing pages where the 10x cost buys content that addresses real fears rather than incremental polish.

Limitations. The evidence is a single instance — one prompt, one domain, one model, each run once — so we cannot separate architectural superiority from stochastic variation or richer-context exploitation a better prompt could capture; this is the primary methodological limitation. We ran no human evaluation: criteria are author-assessed from the ICP, a circularity that prevents independent judgment and, for a user-centred study, is the most significant gap. All agents used the same model; we ran no A/B test of conversion; and a more detailed prompt or a thinner ICP might narrow the gap, since we hypothesise the loop’s contribution scales with persona-document richness.

Confounds and the need for ablation. The independent variable bundles three changes: number of agents, specialised creative roles, and a grounded feedback loop. The R1 comparison suggests multi-agent generation contributes **copy quality and visual consistency** while the feedback loop contributes **content completeness and persona coverage** — but this is observational, not experimental. Four ablations would isolate the components: (1) a single agent with the creative direction pre-loaded; (2) multi-agent generation without customer feedback; (3) a single agent with Self-Refine-style self-evaluation using persona instructions; and (4) the full studio. Until then, attribution of the gains to the feedback loop remains a supported hypothesis, not a proven mechanism.

7 Future Work

We prioritise by expected impact: (1) run the four **ablation** conditions above to isolate each component; (2) a **human evaluation** presenting all three outputs (unlabelled, randomised) to UX designers and persona-matched participants performing a booking task; (3) a low-cost **minimal replication** re-running the skill approach to test whether the content gap is a single-run artifact; (4) a **full replication** (five runs per architecture, two additional domains) reporting mean and variance on persona coverage; (5) **prompt-specificity** experiments; and (6) **cost optimisation** via hybrid architectures that add a single grounded evaluation pass to cheap generation.

8 Conclusion

Holding constant the prompt, model, design system, and reference documents, we found that skill architecture — not model capability, prompt engineering, or design system completeness — was the primary determinant of output quality in this instance. The critical differentiator was not the number of agents or the token volume but **grounded evaluation**: a customer agent that adopts specific personas, scores against a consistent rubric, and ties findings to observable behaviours. The studio’s pre-feedback output shared the single-agent content gaps; the additions emerged only after persona-grounded evaluation flagged them as bounce triggers. For high-stakes, customer-facing deliverables, the 10x investment produced non-linear gains — fundamentally different content, not incremental polish. Methodology prevents bad output, structure produces good decisions, and feedback loops produce the right decisions. Whether this generalises beyond the single domain, model, and prompt tested here requires the proposed replication, ablation, and human evaluation.

9 Ethical Considerations

The studio output contains fabricated medical content: “Dr. Sarah Mitchell, MD” and “Mitchell Orthopaedics” are fictional. “St. David’s Medical Center” is

a real Austin institution; its use in a fictional context is noted as a limitation. The phone number uses the reserved 555 exchange to prevent accidental calls, and the address is fictional. None of the generated pages were deployed, indexed, or made publicly accessible. LLM-generated medical content that blends real and fictional entities risks patient deception if deployed without verification; we recommend mandatory human review before publication, clear labelling of generated content during development, and verification that no real patient could mistake a prototype for an operational practice. The energy cost of 500K+ tokens per page is a further consideration for adoption at scale. The authors declare no competing interests.

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